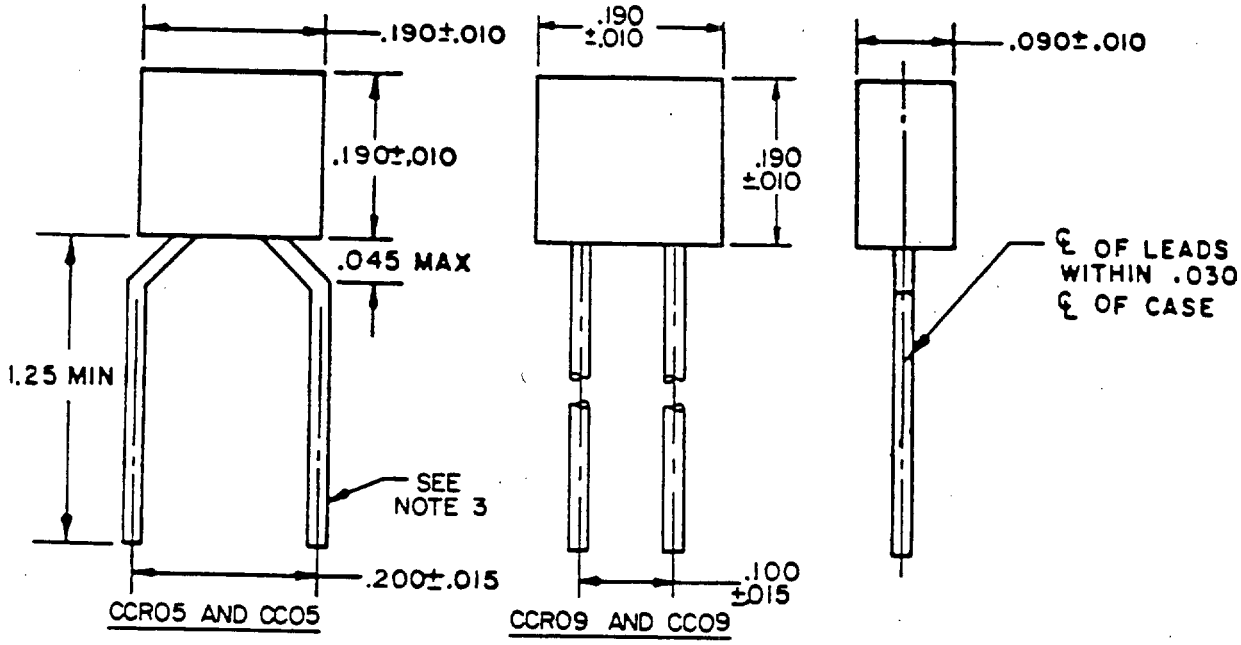
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, CERAMIC DIELECTRIC (TEMPERATURE COMPENSATING), ESTABLISHED AND NON-ESTABLISHED RELIABILITY, STYLES CCR05, CC05, CCR09, AND CC09

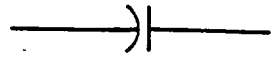
Styles CC05 and CC09 are inactive for design after 19 September 1983. Use CCR05 and CCR09.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the capacitors described herein shall consist of this specification and the latest issue of MIL-C-20.



Configuration without standoff



CIRCUIT DIAGRAM

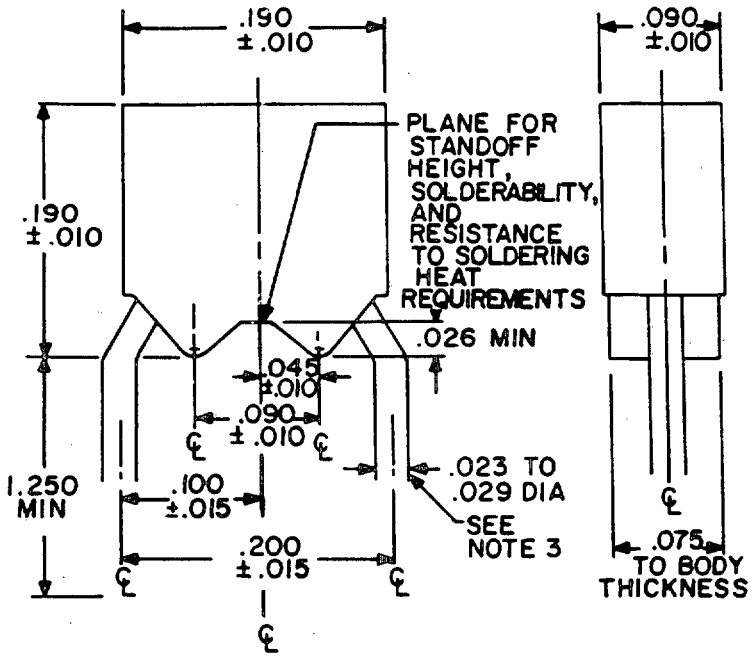
Inches	mm
.001	0.02
.004	0.10
.010	0.25
.015	0.38
.025	0.63
.028	0.71
.030	0.76
.045	1.14
.090	2.29
.100	2.54
.190	4.83
.200	5.08
1.25	31.8

Ⓖ FIGURE 1. Styles CCR05, CC05, CCR09 and CC09 capacitors.

Ⓖ denotes changes

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil - Downloaded: 2026-01-15T17:07Z
Check the source to verify that this is the current version before use.



Optional configuration with standoff (CCR05 only)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Lead diameter shall be .023 (0.58 mm) to .029 (0.74 mm).
4. Optional standoffs for style CCR05 shall be of triangular shape so that the standoffs will provide line contact with surface upon which the capacitor is mounted. The shape of the triangle is optional.
5. Thickness of the standoffs shall be .075 (1.90 mm) to the body thickness.
6. Lead length may be a minimum of .625 (15.88 mm) for use in tape and reel automatic insertion equipment, when specified.

(G) FIGURE 1. Styles CCR05, CC05, CCR09 and CC09 capacitors - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T17:07Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Design and construction:

Ⓖ Dimensions and configuration: See figure 1.

Lead type: Radial.

Case type: Molded.

DC rated voltage: See table I.

Operating temperature range: -55°C to +125°C.

Characteristic: See table I.

Failure rate level (CCR05 and CCR09 only): M (1.0%), P (0.1%), R (.01%) or S (.001%).

Thermal shock and voltage conditioning (CCR05 and CCR09 only): In accordance with MIL-C-20.

Capacitance: Within tolerance specified (see table I).

Dissipation factor: In accordance with MIL-C-20.

Dielectric withstanding voltage: In accordance with MIL-C-20.

Body insulation: Test II.

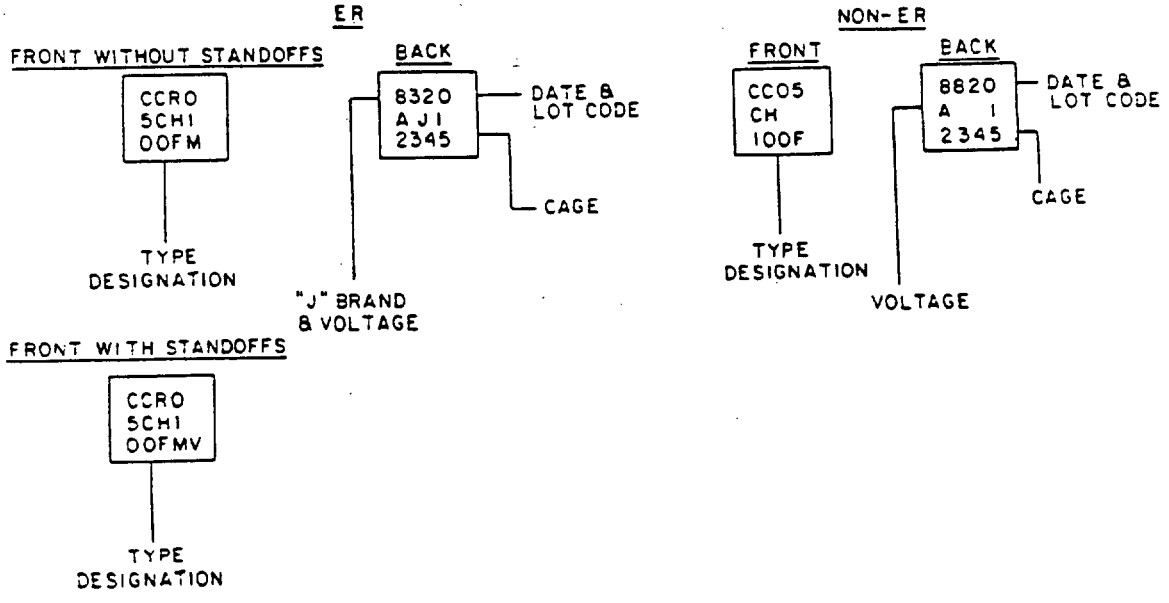
Ⓖ Solderability: For style CCR05 with standoff, the leads shall be solderable to within .024 inch (0.61 mm) from the body egress. For style CCR05 without standoff, solderability shall be in accordance with MIL-C-20.

Ⓖ Resistance to soldering heat: In accordance with method 210 of MIL-STD-202, test condition D.

Ⓖ Workmanship: For style CCR05 with standoff, exposed copper or bare leads shall be permitted to a maximum of .020 inch (0.51 mm) from the body egress. For style CCR05 without standoff, workmanship shall be in accordance with MIL-C-20.

Life: In accordance with MIL-C-20, operating condition 2.

Ⓖ Marking: Method I of MIL-STD-1285. At the option of the manufacturer, the marking may be placed on one side of the capacitor, or the marking may be placed on two lines as long as it is in the same order as shown in the following examples:



Source: https://assist.dla.mil -- Downloaded: 2026-10-15T17:07Z
Check the source to verify that this is the current version before use.

Ⓖ TABLE I. Capacitor characteristics.

Type designation 1/ 2/ 3/	Rated voltage	Nom. cap.	Cap. tol.	Type designation 1/ 2/	Rated voltage	Nom. cap.	Cap tol.
	volts, dc	pF			volts, dc	pF	
CC-0-CX1R0---	200	1.0	BC	CC-0-CH180---	200	18	FGJ
CC-0-CX1R1---	"	1.1	"	CC-0-CG200---	"	20	"
CC-0-CX1R2---	"	1.2	"	CC-0-CG220---	"	22	"
CC-0-CX1R3---	"	1.3	"	CC-0-CG240---	"	24	"
CC-0-CX1R5---	"	1.5	"	CC-0-CG270---	"	27	"
CC-0-CX1R6---	"	1.6	"	CC-0-CG300---	"	30	"
CC-0-CX1R8---	"	1.8	"	CC-0-CG330---	"	33	"
CC-0-CX2R0---	"	2.0	"	CC-0-CG360---	"	36	"
CC-0-CK2R2---	"	2.2	"	CC-0-CG390---	"	39	"
CC-0-CK2R4---	"	2.4	"	CC-0-CG430---	"	43	"
CC-0-CK2R7---	"	2.7	BCD	CC-0-CG470---	"	47	"
CC-0-CK3R0---	"	3.0	"	CC-0-CG510---	"	51	"
CC-0-CK3R3---	"	3.3	"	CC-0-CG560---	"	56	"
CC-0-CK3R6---	"	3.6	"	CC-0-CG620---	"	62	"
CC-0-CK3R9---	"	3.9	"	CC-0-CG680---	"	68	"
CC-0-CJ4R3---	"	4.3	"	CC-0-CG750---	"	75	"
CC-0-CJ4R7---	"	4.7	"	CC-0-CG820---	"	82	"
CC-0-CJ5R1---	"	5.1	"	CC-0-CG910---	"	91	"
CC-0-CJ5R6---	"	5.6	"	CC-0-CG101---	"	100	"
CC-0-CJ6R2---	"	6.2	"	CC-0-CG111---	"	110	"
CC-0-CJ6R8---	"	6.8	"	CC-0-CG121---	"	120	"
CC-0-CJ7R5---	"	7.5	"	CC-0-CG131---	"	130	"
CC-0-CH8R2---	"	8.2	"	CC-0-CG151---	"	150	"
CC-0-CH9R1---	"	9.1	"	CC-0-CG161---	"	160	"
CC-0-CH100---4/	"	10	FGJ	CC-0-CG181---	"	180	"
CC-0-CH110---	"	11	"	CC-0-CG201---	"	200	"
CC-0-CH120---	"	12	"	CC-0-CG221---	"	220	"
CC-0-CH130---	"	13	"	CC-0-CG241---	"	240	"
CC-0-CH150---	"	15	"	CC-0-CG271---	"	270	"
CC-0-CH160---	"	16	"	CC-0-CG301---	"	300	"
CC-0-CG331---	"	330	"	CC-0-CG112---	100	1100	"
CC-0-CG361---	100	360	"	CC-0-CG122---	"	1200	"
CC-0-CG391---	"	390	"	CC-0-CG132---	"	1300	"
CC-0-CG431---	"	430	"	CC-0-CG152---	"	1500	"
CC-0-CG471---	"	470	"	CC-0-CG162---	"	1600	"
CC-0-CG511---	"	510	"	CC-0-CG182---	"	1800	"
CC-0-CG561---	"	560	"	CC-0-CG202---	50	2000	"
CC-0-CG621---	"	620	"	CC-0-CG222---	"	2200	"
CC-0-CG681---	"	680	"	CC-0-CG242---	"	2400	"
CC-0-CG751---	"	750	"	CC-0-CG272---	"	2700	"
CC-0-CG821---	"	820	"	CC-0-CG302---	"	3000	"
CC-0-CG911---	"	910	"	CC-0-CG332---	"	3300	"
CC-0-CG102---	"	1000	"				

See footnotes on next page.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T17:07Z
Check the source to verify that this is the current version before use.

Ⓖ 1/ Complete type designation will include the following:

1st dash: Symbol "R" (for styles CCR05 and CCR09) or dash will be deleted (for styles CC05 and CC09).
2nd dash: Applicable style number (5 for .200 ±.015 lead spacing and 9 for .100 ±0.015 lead spacing).
3rd dash: Applicable capacitance tolerance symbol.
4th dash: Applicable failure rate level symbol (CCR05 and CCR09 only) or dash will be deleted (for styles CC05 and CC09).
5th dash: Symbol "V" (for standoff mounting configuration).

2/ With respect to characteristics in previous issues of this specification sheet: For capacitance values of 2 pF or less, characteristic CX is interchangeable with characteristic CH; for capacitance values of 2.2 pF through 3.9 pF, characteristic CK is interchangeable with characteristic CH; for capacitance values of 4.3 pF through 7.5 pF, characteristic CJ is interchangeable with characteristic CH; and for capacitance values of 11 pF through 18 pF, characteristic CH is interchangeable with characteristic CG. These parts shall be stocked under the same applicable NSN's.

3/ Characteristic CX: Not practically measurable.

4/ Capacitance tolerance: (F) ±1% and (G) ±2%.

Temperature coefficient and capacitance drift: In accordance with MIL-C-20. For capacitors with values of 18 pF or less, the temperature-coefficient tolerance shall be determined in accordance with table II.

TABLE II. Temperature coefficient tolerances.

Permissible capacitance change from capacitance at +25°C in ppm/°C					
Temperature	Characteristic				
	CX	CK	CJ	CH	CG
+125°C	1/	±250 ppm/°C	±120 ppm/°C	±60 ppm/°C	±30 ppm/°C
-55°C 2/	1/	+246.25 -326.25	+116.25 -166.25	+55.00 -91.25	+27.50 -53.75

1/ Not practically measurable.

2/ The ppm/°C values for -55°C were calculated by dividing ppm by negative 80°C.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Navy - OS
Air Force - 99
DLA - ES

User activities:
Navy - AS, MC, SH

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5910-1593)

☆U.S. GOVERNMENT PRINTING OFFICE: 1988 - 505-033/91521

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T17:07Z
Check the source to verify that this is the current version before use.